

HAYAT ULLAH

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EDUCATION

Florida Atlantic University

Boca Raton, FL

PhD Candidate in Computer Science

July 2024 – July 2026

Research Area: “Deep Spatio-Temporal Learning for Video Understanding: Studies in Action Recognition and Temporal Action Localization”

Advisor: Arslan Munir

WORK EXPERIENCE

Florida Atlantic University: ISCAAS Lab

August 2024 – Present

Graduate Research Assistant

Boca Raton, Florida, USA

- Leading a research project entitled “A Multimodal Attention-Based Deep Learning Framework for Real-Time Activity Recognition at the Edge”.
- LLM/VLM:** Performance Benchmarking of Large Language Models and Vision Language Models using distributed training on Air-Cooled and Liquid-Cooled NVIDIA H100 and B200 Nodes.
- VLM:** Developed a codebase for training and finetuning VLMs for spatio-temporal action recognition in videos.
- CV/ML:** Developed a Nano Spatio-Temporal Focal Modulation Network with offline knowledge distillation for efficient human activity recognition on the Edge.
- CV/ML:** Developed a hierarchical multi-stage transformer architecture for context-aware temporal action localization in long untrimmed videos.
- CV/ML:** Created a large-scale sports video dataset for event detection and recognition, and designed tailored prompts to enable key event retrieval and automatic summary generation.
- CV/ML:** Designed and implemented an AI-driven system for detecting and mitigating cyber-attacks in smart grid environments, enabling real-time threat identification, proactive defense mechanisms.
- CV/ML:** Collaborated closely with team members using version control systems such as Git to efficiently manage computer vision code modifications, track project progress, resolve conflicts, and assign tasks, ensuring smooth coordination and streamlined development workflows.

Johnson Controls

June 2026 – August 2026

Data Center AI Intern

Milwaukee, Wisconsin, USA

- LLM/VLM:** Conducted inference and training of state-of-the-art LLMs (Llama 3, Qwen 3, Mistral, and Phi) and VLMs (LLaVA-1.5, Qwen2-VL, and BLIP-2) for AI workload benchmarking and optimization.
- RL/ML:** Developed a reinforcement learning (RL) environment that simulates both IT and OT operations in a liquid-cooled data center.
- LLM/Agents:** Designed an LLM-driven autonomous agent for the RL environment to optimize liquid-cooling operations by reallocating the power leakage from OT side to IT workloads, with the objective of maximizing tokens-per-watt efficiency.

Nine VR

July 2020 – December 2020

Machine Learning Engineer Intern

Seoul, South Korea

- CV/ML:** Proposed and developed Human-Computer Interaction solutions for perceptual quality assessment of panoramic content in VR/AR applications.
- CV/ML:** Developed an Augmented Reality (AR) prototype (utilizing object tracking in immersive contents) for monitoring user experience while using Head Mounted Display (HMD) devices.
- CV/ML:** Developed a computer vision-assisted AR Projection application for Plant Prototypes. Application was delivered to industry for real use, projecting the plant visuals on mock model.

RESEARCH INTERESTS

Video Understanding, Activity Recognition, Temporal Action Localization, Sports Video Analytics, Knowledge Distillation, Adversarial Robustness, Distributed Training (FSDP, Tensor Parallel, Pipeline Parallel, Expert Parallel), VLA Models for Robotics, and LLM and VLM Benchmarking on NVIDIA H100, B200, and B300 GPU Nodes.

SKILLS

Programming Languages: Python, C++, Matlab, Command Line Interface (CLI) scripting.

Python Libraries: PyTorch, Tensorflow, QisKit, OpenCV, Scikit-learn, Numpy, Scikit-image, Matplotlib, Seaborn, and Pandas etc.

AI/ML: Computer Vision, Transformer Neural Networks, VLM’s pretraining and finetuning for videos, Adversarial Robustness, Distributed Training, and Time-Series Classification.

Miscellaneous: Linux, Shell (Bash/Zsh), and Git.